AI and GenAI Assignment - Report

1. Objective

The objective of this assignment is to implement a simple Generative Adversarial Network (GAN) using TensorFlow and the MNIST dataset. The goal is to generate synthetic handwritten digit images that resemble real MNIST digits.

---

2. Brief Explanation of GAN

A Generative Adversarial Network consists of two neural networks:

Generator

- Takes random noise as input.
- Produces synthetic images.
- ***Goal:*** fool the discriminator.

Discriminator

- Receives real and fake images.
- Classifies them as real (1) or fake (0).
- ***Goal:*** correctly detect fake images.

Both networks compete in a **minimax game**:

- Generator improves to fool discriminator.
- Discriminator improves to detect fakes.

---

3. Architecture Summary

Generator

Layers:

1. Dense layer
2. Batch Normalization
3. LeakyReLU
4. Conv2DTranspose layers
5. Output with Tanh activation

Output: 28×28 grayscale image.

Discriminator

Layers:

1. Conv2D
2. LeakyReLU
3. Dropout
4. Flatten
5. Dense layer with sigmoid

Output: Probability (real or fake).

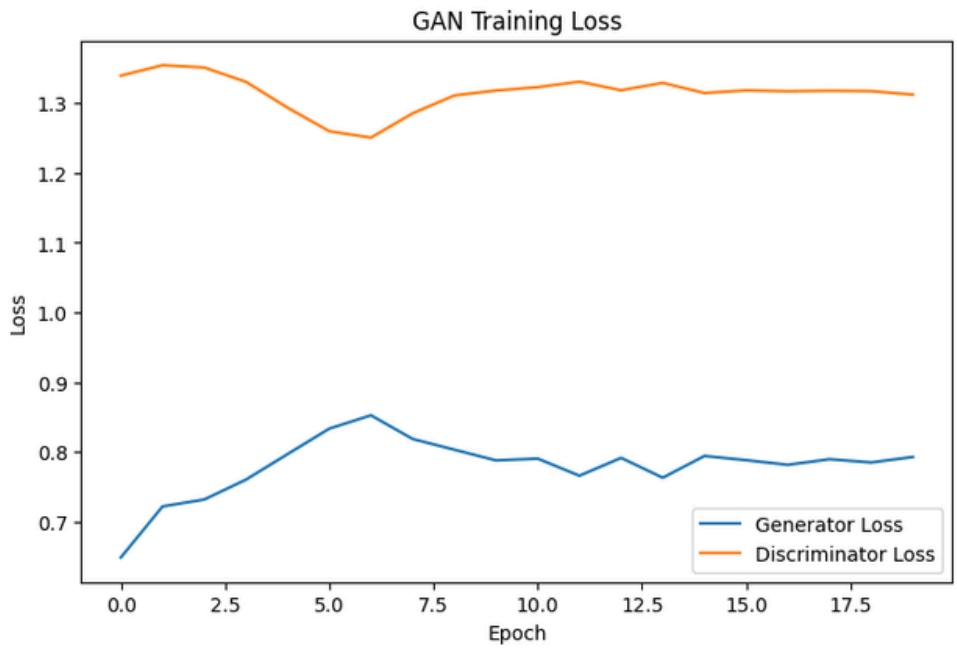
---

4. Training Details

- Dataset: MNIST
- Normalization: [-1, 1]
- Loss: Binary Cross-Entropy
- Optimizer: Adam (lr=0.0002, β_1=0.5)
- Epochs: 20

---

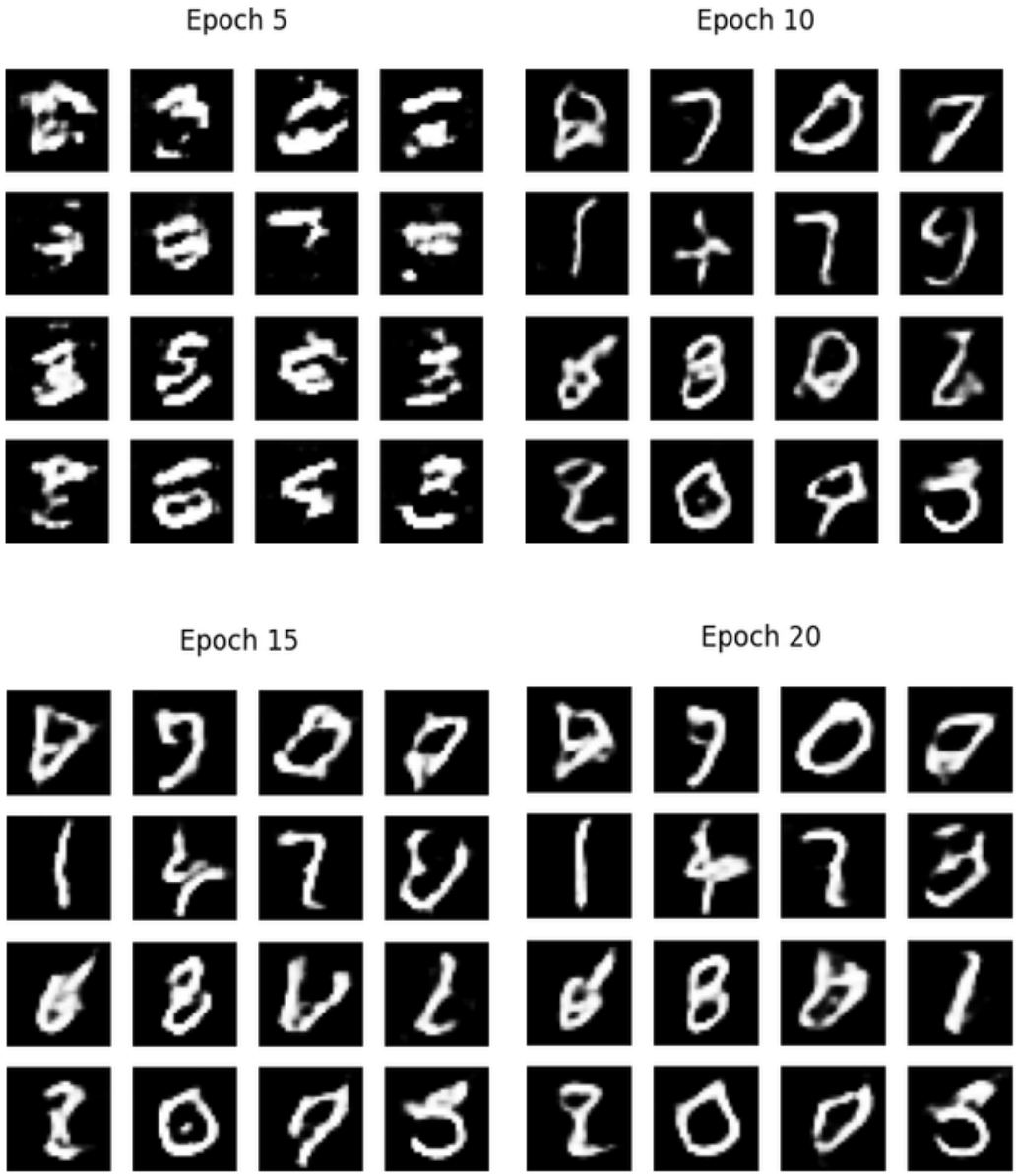
5. Loss Graph



Observation:

The graph shows that both generator and discriminator losses change over the training epochs instead of decreasing smoothly. This is normal in GAN training because both models learn in competition with each other. At the beginning, the generator loss increases, which means the generator is still learning and not producing good outputs. After a few epochs, the generator loss starts to decrease and becomes stable around 0.78–0.80, showing improvement in generating more realistic images. The discriminator loss slightly decreases at first and then stabilizes around 1.30–1.32. This indicates that the discriminator is no longer easily able to distinguish between real and fake images. Overall, both losses become stable with small fluctuations, which suggests that the GAN has reached a balanced state. This means the generator and discriminator are learning effectively, and the model is performing well without major instability.

6. Sample Generated Images



Observation:

- Early epochs: noisy shapes.
- Later epochs: clearer digit structures.
- The generator gradually improves realism.

---

7. Challenges Faced

- Training instability in GANs.
- Loss values fluctuate instead of decreasing.
- Balancing generator and discriminator learning rates.

---

8. Conclusion

The GAN successfully generated handwritten digit images similar to the MNIST dataset. The adversarial training helped the generator improve over epochs, producing increasingly realistic digits.